CSE 400 Lecture Scribe

L5_A_S2.pdf
PDF

Make an exam-oriented lecture scribe for my course CSE 400: Fundamentals of Probability in Computing. You can only use the provided lecture slides and attached material (if any) with this prompt. Do NOT add any material, explanations, examples, interpretations, or methods beyond what is present in the provided material. Follow the lecture slide order exactly. Do not interpret anything yourself. Mention all definitions, notation, assumptions, proofs, derivations, and worked examples step by step as presented in the lecture. And mark every heading and subheadings clearly of the definitions and examples. Make sure to maintain a formal academic tone suitable for exams revision. Also, do not give examples outside of the material, include all the intermediate steps and make sure the scribe is not out of the provided material.

Got it, Abhi. I will do this **exactly exam-style**, **strictly limited to the provided Lecture 5 slides only**, with **no extra explanation or outside examples**, and in the **exact slide order**.

Below is your **formal exam-oriented lecture scribe** for:

CSE 400 — Fundamentals of Probability in Computing
Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function
(Only from the provided slides)

CSE 400: Fundamentals of Probability in Computing

Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

Dhaval Patel, PhD — January 20, 2026

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1. Bayes' Theorem

1.1 Weighted Average of Conditional Probabilities

Event Decomposition

Let A and B be events.

We may express:

$$A = AB \cup AB^c$$

Reason:
For an outcome to be in A, it must either:

- be in both A and B, or
- be in A but not in B

Since AB and AB^c are mutually exclusive, by **Axiom 3**:

$$Pr(A) = Pr(AB) + Pr(AB^c)$$

Using conditional probability:

$$Pr(AB) = Pr(A \mid B)Pr(B)$$

$$Pr(AB^c) = Pr(A \mid B^c)Pr(B^c)$$

Thus:

$$Pr(A) = Pr(A \mid B)Pr(B) + Pr(A \mid B^c)[1 - Pr(B)]$$

Key Statement

The probability of event A is a **weighted average** of the conditional probabilities, with weights given as the probability of the event on which it is conditioned.

1.2 Learning by Example

EXAMPLE #3.1 (Part 1/2)

An insurance company divides people into two classes:

- Accident prone
- Not accident prone

Statistics:

- Accident-prone person has an accident within 1 year with probability:

$$0.4$$

- Not accident-prone person has an accident within 1 year with probability:

$$0.2$$

Assume:

$$Pr(A) = 0.3$$

where A = event that the policyholder is accident prone.

Let:

- A_1 = event that the policyholder has an accident within 1 year

We want:

$$Pr(A_1)$$

Solution
Condition on whether the policyholder is accident prone:

$$Pr(A_1) = Pr(A_1 \mid A)Pr(A) + Pr(A_1 \mid A^c)Pr(A^c)$$

Substitute values:

$$Pr(A_1) = (0.4)(0.3) + (0.2)(0.7)$$

$$= 0.12 + 0.14 = 0.26$$

So:

$$Pr(A_1) = 0.26$$

EXAMPLE #3.1 (Part 2/2)

Suppose a new policyholder has an accident within a year.

What is:

$$Pr(A \mid A_1)?$$

Solution

$$Pr(A \mid A_1) = \frac{Pr(AA_1)}{Pr(A_1)}$$

$$= \frac{Pr(A)Pr(A_1 \mid A)}{Pr(A_1)}$$

Substitute:

$$= \frac{(0.3)(0.4)}{0.26}$$

$$= \frac{0.12}{0.26} = \frac{6}{13}$$

Thus:

$$Pr(A \mid A_1) = \frac{6}{13}$$

1.3 Formal Introduction: Law of Total Probability and Bayes Theorem

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Law of Total Probability (Formula 3.4)

Suppose:

$$B_1, B_2, \ldots, B_n$$

are mutually exclusive events such that:

$$\bigcup_{i=1}^{n} B_i = B$$

Then:

$$Pr(A) = \sum_{i=1}^{n} Pr(AB_i)$$

Since:

$$Pr(AB_i) = Pr(A \mid B_i)Pr(B_i)$$

We obtain:

$$Pr(A) = \sum_{i=1}^{n} Pr(A \mid B_i)Pr(B_i)$$

This is the **Law of Total Probability**.

Bayes Formula (Proposition 3.1)

Using:

$$Pr(AB_i) = Pr(B_i \mid A)Pr(A)$$

We get:

$$Pr(B_i \mid A) = \frac{Pr(A \mid B_i)Pr(B_i)}{\sum_{j=1}^{n} Pr(A \mid B_j)Pr(B_j)}$$

This is the **Bayes Formula**.

Definitions

- $Pr(B_i)$ is the **apriori probability**
  (formed from self-evident or presupposed models)
- $Pr(B_i \mid A)$ is the **posteriori probability**
  (derived after observing certain events)

1.4 Learning by Example

EXAMPLE #3.2 (Cards Problem)

There are 3 cards:

1. Card 1: Red–Red
2. Card 2: Black–Black
3. Card 3: Red–Black

One card is randomly selected and placed down.

Given:

Upper side is red.

Find probability that the other side is black.

Solution

Let:

- RR: chosen card is all red
- BB: chosen card is all black
- RB: chosen card is red–black
- R: event that upturned side is red

We want:

$$Pr(RB \mid R)$$

$$Pr(RB \mid R) = \frac{Pr(RB \cap R)}{Pr(R)}$$

$$= \frac{Pr(R \mid RB)Pr(RB)}{Pr(R \mid RR)Pr(RR) + Pr(R \mid RB)Pr(RB) + Pr(R \mid BB)Pr(BB)}$$

Substitute:

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)}$$

$$= \frac{1/6}{1/3 + 1/6}$$

$$= \frac{1/6}{1/2} = \frac{1}{3}$$

Thus:

$$Pr(RB \mid R) = \frac{1}{3}$$

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2. Random Variables

2.1 Motivation and Concept

When an experiment is performed, we are often interested in some function of the outcome.

Examples:

- Dice tossing: focus on the sum
- Coin flipping: focus on number of heads

These real-valued functions are called **Random Variables**.

Definition

A random variable X on sample space Ω is a function:

$$X : \Omega \to \mathrm{R}$$

assigning each sample point $\omega \in \Omega$ a real number $X(\omega)$.

Distribution

Important aspects:

1. The set of values it can take
2. The probabilities with which it takes those values

Event notation:

$$\{\omega \in \Omega : X(\omega) = a\}$$

Abbreviated as:

$$X = a$$

Probability:

$$Pr[X = a]$$

The collection of these probabilities is called the **distribution** of X.

Visualization

The distribution can be shown as a bar diagram:

- x-axis: values of X
- bar height at a: $Pr[X = a]$

2.2 Random Variable Example

Example 1: Tossing 3 Fair Coins

Let Y denote number of heads.

Possible values:

$$0, 1, 2, 3$$

Probabilities:

$$P(Y = 0) = P(t, t, t) = \frac{1}{8}$$

$$P(Y = 1) = P(t, t, h), (t, h, t), (h, t, t) = \frac{3}{8}$$

$$P(Y = 2) = P(t, h, h), (h, t, h), (h, h, t) = \frac{3}{8}$$

$$P(Y = 3) = P(h, h, h) = \frac{1}{8}$$

Since Y must take one of these values:

$$1 = \sum_{i=0}^{3} P(Y = i)$$

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3. Probability Mass Function (PMF)

3.1 Concept

A random variable that can take on at most a countable number of values is said to be **discrete**.

Let X be discrete with range:

$$R_X = x_1, x_2, x_3, \ldots$$

The function:

$$p(x_k) = Pr(X = x_k)$$

is called the **Probability Mass Function (PMF)** of X.

Since X must take one of the values x_k:

$$\sum_k p(x_k) = 1$$

3.2 PMF Example

The PMF is given by:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

Find:

$$P(X = 0)$$

and

$$P(X > 2)$$

Solution

Since:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

But:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So:

$$ce^{\lambda} = 1$$

$$c = e^{-\lambda}$$

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(1) Finding $P(X = 0)$

$$P(X = 0) = p(0) = c\frac{\lambda^0}{0!}$$

$$= e^{-\lambda}$$

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(2) Finding $P(X > 2)$

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

Class Participation

Switch to Campuswire for discussion.

✅ Lecture 5 Scribe Complete

Abhi, this is a **strict slide-faithful exam scribe** with:

- All definitions and formulas
- All worked examples step-by-step
- No extra interpretation
- Exact lecture order

If you want, I can generate the same style scribe for **Lecture 6** next.